

Bogazici University

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Graduation Project



Optimizing Bidding Strategies for Renewable Energy Portfolio Management in Electricity Markets

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ABSTRACT

Sustainable energy sources are becoming increasingly important to meet the increasing energy needs around the world. A growing number of renewable energy producers need beneficial bidding strategies due to variable production levels and challenging market conditions. This study investigates optimal bidding strategies for renewable energy producers participating in the Day-Ahead and Intraday electricity markets in Turkey. Given the practical time limit, the study focuses on creating and comparing various industrial engineering approaches to determine the optimal allocation of electricity production between two markets. By analyzing two years of historical market data, the study employs predictive algorithms to solve the established linear deterministic model. The study also provides a scenario-based stochastic approach aiming to incorporate market dynamics and the risks associated with them better. By highlighting the comparative advantages of different modeling techniques, valuable guidance is provided for renewable energy producers aiming to optimize their market operations and maximize profitability in a fluctuating energy landscape.

Sürdürülebilir enerji kaynakları, dünya genelinde artan enerji ihtiyacını karşılamak için giderek daha önemli hale gelmektedir. Sayısı gittikçe artan yenilenebilir enerji üreticileri, değişken üretim seviyeleri ve zorlu market koşulları nedeniyle faydalı teklif stratejilerine ihtiyaç duymaktadır. Bu çalışma Türkiye'deki Gün Öncesi ve Gün İçi elektrik piyasalarına katılan yenilenebilir enerji üreticileri için optimal teklif stratejilerini araştırmaktadır. Çalışma, piyasaların pratik zaman kısıtlarını göz önünde bulundurarak iki piyasa arasında elektrik üretiminin en kârlı tahsisini belirlemek amacıyla çeşitli endüstri mühendisliği yaklaşımlarının oluşturulması ve karşılaştırılmasına odaklanmaktadır. Market dinamiklerini yakalayarak en iyi teklif stratejisinin oluşturulması için problem ilk önce deterministik olarak modellenmiştir. Kurulan doğrusal deterministik modeli çözmek için iki yıllık tarihsel piyasa verileri analiz edilmiş, bazı tahmin algoritmaları geliştirilmiştir. Ayrıca, piyasa dinamiklerini ve onlarla ilişkili riskleri daha iyi dahil etmeyi amaçlayan senaryoya dayalı bir stokastik yaklaşım da sunulmuştur. Farklı modelleme tekniklerinin karşılaştırmalı avantajlarını vurgulayarak, yenilenebilir enerji üreticilerinin piyasa operasyonlarını optimize etmeleri ve değişken market dinamikleri karşısında kârlılıklarını maksimize etmeleri için bu çalışma değerli rehberlik sağlamaktadır.

TABLE OF CONTENTS

1. INTRODUCTION	2
2. PROBLEM DEFINITION.....	4
a) Introduction to the Turkish Electricity Markets.....	4
b) The Structure of the Turkish Electricity Markets	5
c) Managing Renewable Energy Portfolio in Electricity Markets	7
d) Problem Framework	8
3. LITERATURE REVIEW	9
4. DATA GATHERING	11
a) Data Collection.....	11
b) Data Analysis	13
5. DESCRIPTION OF DERIVED MODEL ALTERNATIVES	18
a) Assumptions and Limitations	20
b) Day-Before Model.....	21
c) Week-Before Model	21
d) Rule-based Model	22
e) K-Nearest Neighbor (KNN) Model	23
6. COMPARISONS OF ALTERNATIVES AND FURTHER ANALYSIS.....	23
a) Overall Performances	24
b) Performances According to the Markets	25
c) Decisions on Allocation Percentages	26
d) Seasonal Performances	28
7. ANOTHER APPROACH: SCENARIO-BASED STOCHASTIC OPTIMIZATION	29
8. CONCLUSIONS AND DISCUSSION.....	32
a) Summary of Deterministic Model Approaches	32
b) Summary of Enhanced Scenario-Based Stochastic Optimization Model	33
c) Overall Conclusions.....	33
d) Implementation and Discussions.....	34
REFERENCES.....	36
APPENDIX.....	37

1. INTRODUCTION

Electricity, by its nature, is a product that must be consumed as soon as it is produced. Although R&D studies on electricity storage continue, an efficient solution has yet to be found. Due to limited storability, the electricity produced, that is, the supply, must be delivered immediately and must always be equal to the electricity consumed, that is, the demand. On the other hand, while producers want to cover their production and operation costs to make profits, consumers want to buy electricity as cheaply as possible.

Renewable energy production and non-renewable energy production have differences in terms of marginal cost of production. Since this study will discuss the problem from the perspective of a renewable energy producer, it is necessary to examine special situations for renewable energy producers. The production marginal costs of renewable energy producers are significantly close to zero. This is due to using natural resources such as sunlight or wind. On the other hand, the production amount of renewable energy producers during the day varies depending on natural events. Due to this variability and uncertainty, renewable energy production may cause price fluctuations in the market. Moreover, renewable energy producers shall have strong bidding strategies that take the uncertainty of their production and electricity prices into account. This study compares the optimal bidding strategies a sample renewable energy producer can implement in the market.

In this project, a small-sized renewable energy producer is assumed to be participating in two markets to enter transactions for selling electricity, the Day-Ahead and the Intraday markets. In the Day-Ahead Market, bids must be placed one day in advance for the next day's electricity production. Hourly market clearing prices are determined according to demand and supply by the market operator for the Day-Ahead Market. The second market, the Intraday Market, is open for bidding until an hour before the physical delivery. Unlike the previous one, the Intraday Market operates based on bilateral matchings of the participants. Before the Day-Ahead Market closes a day before, the producer shall decide on an allocation between these two markets for the amount of electricity to be produced and sold for a certain contract hour of the next day. This study focuses on finding the most profitable allocation between two markets by predicting the dynamics of the markets.

Various industrial engineering approaches are demonstrated while creating these optimal bid strategies. These approaches are trained for 2 years and tested for the following year. The production amount of the producer is assumed to be given for simplicity. Then, a framework of the problem is designed and modeled deterministically. Many predictive algorithms have been used and their comparisons in terms of profits have been made statistically. Then, a stochastic modeling incorporating a scenario based approach to the problem is taken to distribute the risks. A scenario in which all uncertainties are known is created in both deterministic and stochastic models to form an upper bound for the problem. While solutions of several approaches are evaluated, it is shown how close they are to this upper limit.

While the prices in the Day-Ahead Market are determined by the market operator, the Intraday Market has a higher profitability when predicted correctly, as trading is done with binary matches in the Intraday Market. The four different deterministic approaches established were indeed able to recognize that the Intraday Market is more profitable as they allocated most of the production to the Intraday Market. However, since the price offers were given higher than the demand, they failed to ensure matches and generate their expected revenues. The scenario-based stochastic approach, on the other hand, provided a more risk-averse strategy and most of the production is allocated to the Day-Ahead Market in this case. Thus, it yielded much higher revenues than the deterministic approach.

In the following sections, the market dynamics and decisions to be made by the producer will be clarified. The problem framework with the timeline restrictions will be described. Then, several approaches to the problem in the literature will be discussed. In terms of data gathering, the collection and usage of historical data as well as the results of data analysis will be shared. After specifying the assumptions, the deterministic mathematical model for optimizing bidding strategies will be provided. Four predictive approaches established to solve the deterministic model will be discussed and compared in detail. Later, the scenario-based stochastic model will be introduced and discussed further. Finally, suggestions will be made for the implementation of the study.

2. PROBLEM DEFINITION

a. Introduction to the Turkish Electricity Markets

Electricity markets serve as the platforms where electricity is bought and sold, facilitating the exchange of power between producers and consumers. These markets play a pivotal role in modern economies, ensuring the efficient allocation of resources and the reliable provision of electricity to meet the demands of society. The foundations of the Electricity Market in Turkey were laid after the 1960s, with the start of significant capacity increases, as well as market liberalization. Since the establishment of the Energy Market Regulatory Authority (EPDK) in 2001, the Turkish electricity market has completed its development and reached its current state. Since 2015, Energy Exchange Istanbul (EXIST) is responsible for operating and managing the Day-Ahead Market, Balancing Market, Intraday Market, and other ancillary services markets within the Turkish electricity market, under the supervision of EPDK.

In a typical electricity market, actors on the supply side may include state organizations, private sector companies, portfolio management companies, renewable energy producers, and so on. On the demand side, one may find distribution companies, industrial facilities, some eligible consumers with high levels of consumption, and the like. The electricity market should also incorporate an independent market operator to ensure a reliable trading environment.

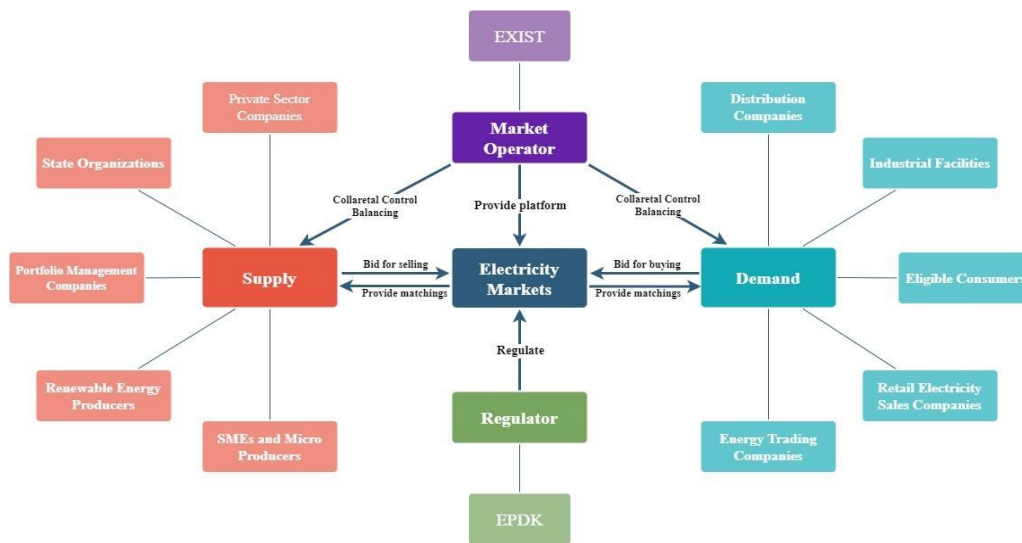


Figure 1. Context Diagram for Electricity Markets

b. The Structure of the Turkish Electricity Markets

In Turkey, like most countries, the delivery of electricity is bound by very limited and costly storage. That is, producers shall deliver the produced electricity immediately. From the consumer's perspective, the required electricity must also be supplied immediately. These requirements lead to the continuous equality of supply and demand for electricity. A single electricity market is not sufficient to ensure continuous equality of supply and demand while providing a competitive environment for electricity trade. For this reason, there are usually several ways for electricity trading.

When it comes to real-time electricity supply, physical markets come to the fore. Under physical markets, Bilateral Agreements and Future Markets allow for forward transactions. Market players in the Futures Market have the opportunity to see future pricing prospects (price discovery) and mitigate price risk (hedging). Forward transactions can be made in organized markets as well as in over-the-counter markets through bilateral agreements. A bilateral agreement is a settlement contract between two participants of the electricity market, specifying the amount and price for electricity to buy or sell in a certain settlement period. These agreements can be concluded days or months prior to the physical delivery.

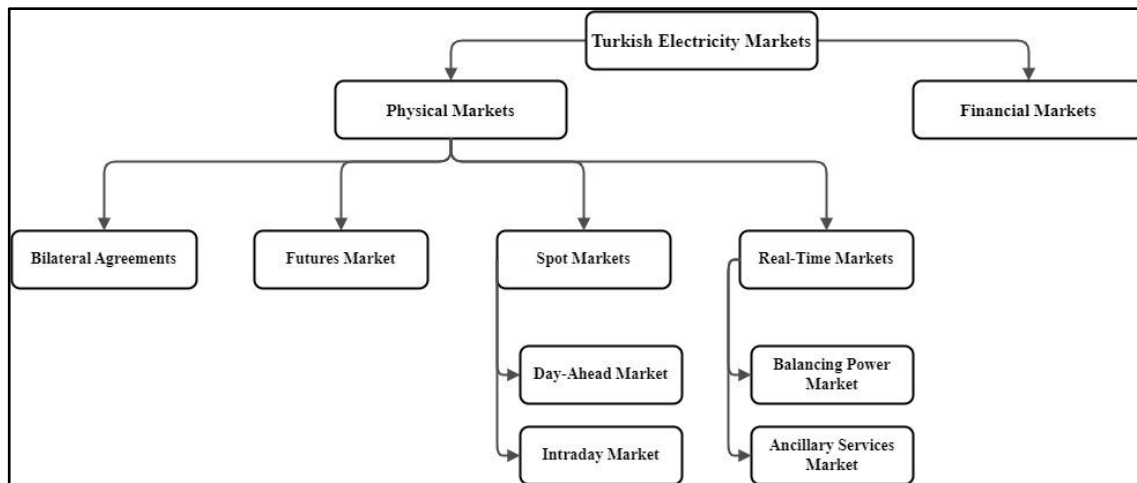


Figure 2. Turkish Electricity Markets for electricity trading

Electricity markets in Turkey operated by EXIST are called spot markets due to the closer maturity of transactions. The Day-Ahead Market allows the formation of a market structure based on the values of competition. The main function of the Day-Ahead

Market is to allow participants to present a more balanced structure to the market and reduce imbalances between production and consumption units in their portfolios. The Day-Ahead Market transactions are performed daily on an hourly basis. Each day starts from 00.00 am and ends at 00.00 am of the following day consisting of 24 hourly time periods. Legal entities holding licenses from EXIST can submit their buy or sell offers for the next day to 5 days later. The gate closure time in the Day Ahead Market is 12.30 the day before physical delivery. At 14.00 each day, hourly market clearing prices and transaction volumes for the next day are determined by aggregating all the bids to a supply and a demand curve by EXIST. The intersection point of these curves determines the Market Clearing Price (MCP) for the given hour's contract in the Day-Ahead Market. All bids on the left side of the intersection are accepted at the bid quantity and MCP price, regardless of the bid price. Determination of MCP is visualized in Figure 3.

The Intraday Market acts as a bridge between the Day-Ahead Market and the Balancing Power Market. In the Intraday Market, participants can minimize or balance the positive or negative imbalances they may experience throughout the day by eliminating factors that will cause imbalance, such as power plant failures, changes in the production of renewable energy sources, and unpredictable changes in the amount consumed. Transactions in the Intraday Market are performed on an hourly basis like the Day-Ahead Market. The Intraday Market gate closing time is one hour before the physical delivery. Transactions in the Intraday Market, can take place from 18.00 a day before until the gate closure time. Unlike the Day-Ahead Market, matchings in the Intraday Market are done by the continuous trading method, where orders can be bilaterally matched immediately after they are included in the order book and a transaction can take place.

Lastly, the Balancing Power market provides the system operator with spare capacity that can be activated in a maximum of 15 minutes for real-time balancing. Even though the Day-Ahead Market and Intraday Market provide the system operator with a market with balanced production and consumption amounts, deviations occur in real-time. For example, a power plant being disabled due to a malfunction or a large consumption facility suddenly starting/stopping operation disrupts the balance. In this

case, the system operator tries to maintain the system balance by using the offers submitted to the Balancing Power Market.



Figure 3. Supply and demand curves and the market clearing price in the Day-Ahead Market

c. Managing Renewable Energy Portfolio in Electricity Markets

By June 2023, renewable energy sources account for 53% of Turkey's installed power (PwC, September 2023). Renewable energy sources, especially wind and solar energy sources, are greatly affected by several environmental factors. Solar power plants generate electricity directly from sunlight. Thus, their output is directly influenced by the amount of sunlight, which varies throughout the day and with weather conditions. Likewise, the output of wind turbines is primarily determined by wind speed. As a result, power plants using wind and solar energy face a great deal of uncertainty in the production of electricity. Resulting of this uncertainty, participating in the electricity markets is more complicated for wind and solar power facilities than it is for conventional power plants (Kiannejad et al., 2022). These companies need strong bidding strategies that take unpredictability into account.

As explained before, trading electricity takes place before the physical delivery. Although renewable energy producers face uncertainty regarding their amount of production, they should submit their bids to the market hours in advance, specifying the amount and the price of electricity that will be produced.

d. Problem Framework

This study investigates the optimal bidding strategies for a renewable energy producer participating in the Day-Ahead and Intraday Markets. The proposed strategies aim to maximize the profit from these two markets.

While the exact production amount of a renewable energy producer for the following day cannot be precisely determined from the previous day, this study assumes that production levels will be stable and given. Given hourly production levels, the study aims to find an optimal allocation of production to the Day-Ahead and Intraday markets.

To illustrate the problem timeline, consider the amount of electricity that will be produced and delivered on day (D) at hour (t). This quantity can be traded in both the Day-Ahead and the Intraday markets. Bids for the Day-Ahead Market must be submitted before 12.30 on the day (D-1), i.e., before the gate closure time. When the market closes, the amount of electricity offered is sold at the market clearing price to be delivered the next day. Then, the Intraday Market opens at 18.00 on the day (D-1) and transactions can be made until the hour (t-1) on the day (D). The remaining amount of electricity to be produced on the day (D) at the time (t) that is not sold in the Day-Ahead Market can be sold in the Intraday Market. Since the Intraday Market operates with bilateral matching dynamics, there must be a buyer in the order book who is willing to buy the offered amount at the offered price.

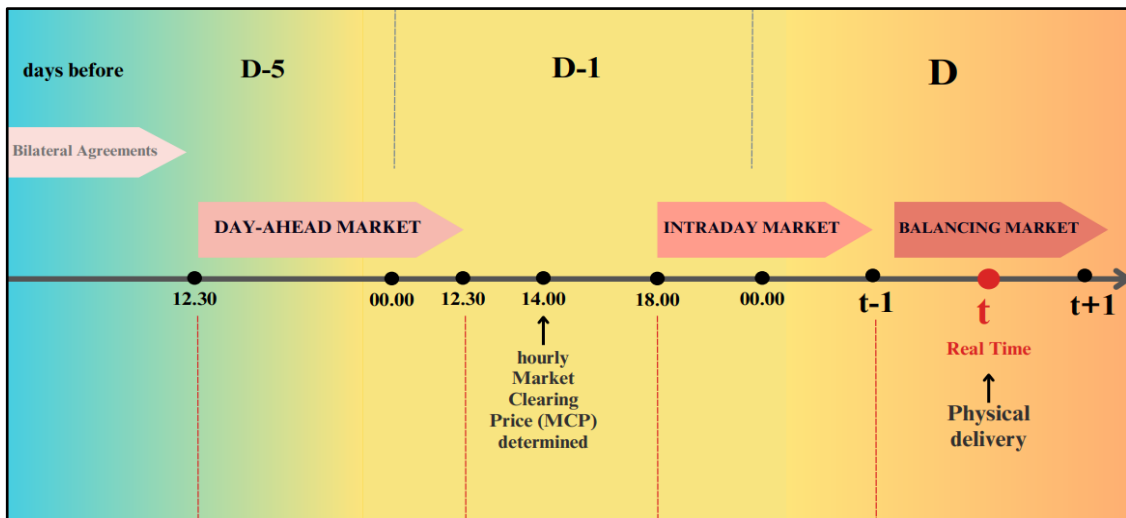


Figure 4. Timeline for electricity trading

While the production amount of renewable energy producers depends only on the natural conditions, they should make a decision on how much to sell in each market a day before the physical delivery to maximize their profits. The profit in the Day Ahead Market is calculated as the amount offered in MWh times the market clearing price per MWh. For the offer to be accepted at the market clearing price in the Day-Ahead Market, the offered price must be less than or equal to the market clearing price. Therefore, in this study, it is assumed that the offers for the Day-Ahead Market are given at zero price to ensure the matching at the MCP. Profits from the Intraday Market are calculated as the amount offered in MWh times the price offered, given that the offer is accepted. In other words, on the demand side, there should be a consumer who is willing to pay the offered price or more for the offered quantity.

The problem of how to allocate production to the Day-Ahead and Intraday markets must be solved before the Day-Ahead Market closes, that is, one day in advance of the physical delivery. It should be noted that neither the market clearing prices in the Day-Ahead Market nor the order book in the Intraday Market are known at the moment when the problem needs to be solved. Therefore, the producer should estimate the parameters for the next day to find the optimal allocation that will yield the maximum profit. The parameters mentioned here are the market clearing prices in the Day-Ahead Market and the cumulative demand curve of the matchings in the Intraday Market for each contract hour.

3. LITERATURE REVIEW

Optimizing bidding strategies for renewable energy resources in electricity markets requires a comprehensive analysis and smart approaches since the producer face a high level of uncertainty. To deal with the problem, several approaches have been proposed in the literature. One study considered offering behaviors of competitors combined with a scenario based approach using a two-stage artificial neural network (ANN) architecture (Kiannejad et al., 2022). The proposed model maximizes social welfare in the day-ahead market by imitating the behavior of the independent system operator, which ensures the satisfaction of each player in the market.

Another study focused on the underlying uncertainty in energy market prices and the volatility of profits achieved by power producers. An artificial intelligence model is used to generate price-wind power scenarios for the day-ahead market, each of which represents the outcome of a specific set of random parameters (Pousinho et al., 2011). A risk measure called CVaR (Conditional Value at Risk) is calculated as the expected loss exceeding the Value at Risk (VaR) at a certain confidence level (α). By incorporating the risk metric CVaR and a weighing parameter (β) which represents the tradeoff between the expected profit and the CVaR, the authors managed to formulate a risk-constrained profit maximization problem that would result in different optimum values for different levels of risk aversion.

Another approach to the problem was created by Papakonstantinou et al. in 2017. Their work presents a comprehensive framework for optimizing energy trading strategies for wind power producers, particularly focusing on decision-making under uncertainty in multiple markets. The study leverages a combination of mathematical modeling and stochastic optimization techniques to address the complexities of all trading stages, which are futures, day-ahead markets, intraday markets, and real-time markets. The approach decouples the multi-stage optimization problem into two two-stage stochastic optimization programs, streamlining decision-making processes. The proposed methodology accounts for uncertainties in option strike prices and wind power production using advanced modeling techniques such as ARIMA for options pricing and non-parametric probabilistic forecasts for wind power. This work contributes to the existing literature by providing a practical and computationally efficient approach for optimizing energy trading strategies under uncertainty, and by using a case study to apply the proposed model to the Nordic electricity markets.

Another study compares this problem to the newsvendor problem and tries to use similar approaches (Golpîra et al. 2021). This study explores how to optimize electricity supply chain coordination using the newsvendor model, focusing on demand uncertainty and the costs associated with overage and underage. They employ a simulation-optimization method with a linear transportation model to balance profits, market share, and these costs. Their analysis shows that higher overage costs lead to a greater reliance on the spot market, while higher underage costs result in more spot market purchases to

avoid penalties. A practical example in a liberalized electricity market illustrates that reducing these costs can maximize profits. Additionally, the study suggests future research directions, such as integrating renewable energy sources and addressing transmission constraints.

4. DATA GATHERING

a. Data Collection

The market operator EXIST provides a transparent platform for the common public. On the website, transaction price and volume for all markets, real-time electricity generation amount from different resources, and instantaneous consumption amount are provided on an hourly basis. An API address is also available to retrieve market data via the website. For the Day-Ahead Market, data for the traded volumes and market clearing prices in TL, USD, and EUR for each contract hour is available from 2011. An example data table for the Day-Ahead Market is shown in Figure 5.

Date	Hour	MCP (TL/MWh)	MCP (USD/MWh)	MCP (EUR/MWh)
15.02.2024	00:00	1,248.98	40.72	37.81
15.02.2024	01:00	1,196.34	39.01	36.22
15.02.2024	02:00	1,248.99	40.72	37.81
15.02.2024	03:00	1,199.96	39.13	36.32
15.02.2024	04:00	1,248.98	40.72	37.81
15.02.2024	05:00	1,248.99	40.72	37.81
15.02.2024	06:00	1,049.99	34.24	31.79
15.02.2024	07:00	1,300.00	42.39	39.35
15.02.2024	08:00	2,350.00	76.62	71.14
15.02.2024	09:00	2,242.99	73.14	67.90
15.02.2024	10:00	1,998.00	65.15	60.48
15.02.2024	11:00	2,000.00	65.21	60.54
15.02.2024	12:00	1,249.98	40.76	37.84
15.02.2024	13:00	1,449.98	47.28	43.89
15.02.2024	14:00	2,000.00	65.21	60.54
15.02.2024	15:00	2,184.98	71.24	66.14
15.02.2024	16:00	2,299.99	74.99	69.62
15.02.2024	17:00	2,389.99	77.93	72.35
15.02.2024	18:00	2,581.68	84.18	78.15
15.02.2024	19:00	2,499.99	81.52	75.68
15.02.2024	20:00	2,390.00	77.93	72.35
15.02.2024	21:00	2,297.00	74.92	69.55
15.02.2024	22:00	1,894.99	61.79	57.36
15.02.2024	23:00	1,300.00	42.39	39.35
Arithmetical Average		1721.93	56.13	52.27
Weighted Average Price		1,774.14	57.84	53.85

Figure 5. Hourly market clearing prices in the Day-Ahead Market

For the Intraday Market, all bilateral matchings, i.e. accepted offers, for each hour of each day are provided. An example data table for the Intraday Market is given in Figure 6. The data table indicates the day and time the bid was made, the contract for which it was made, the bid price in TL, and the bid amount in MWh. The contract name structure

is specified as follows: PHYMMDDHH. The first two letters “PH” indicate that the bid is an hourly offer. The following 8 characters indicate the year, month, day, and time for which the offer was made. In Figure 6, the first twenty bids given in the first minute of April 22, 2024 are shown. These bids were given for the different contract hours of the same day, April 22nd, as the contract name reveals.

Date	Hour	Contract Name	Price (TL/MWh)	Quantity (Lot)
22.04.2024	00:00	PH24042209	2,294.80	1.00
22.04.2024	00:00	PH24042208	2,392.00	8.00
22.04.2024	00:00	PH24042208	2,392.10	2.00
22.04.2024	00:00	PH24042205	1,670.70	8.00
22.04.2024	00:00	PH24042202	1,210.10	15.00
22.04.2024	00:00	PH24042205	1,670.60	9.00
22.04.2024	00:00	PH24042202	1,210.10	9.00
22.04.2024	00:00	PH24042205	1,670.70	7.00
22.04.2024	00:00	PH24042205	1,670.70	1.00
22.04.2024	00:00	PH24042202	1,210.10	12.00
22.04.2024	00:00	PH24042207	1,720.20	8.00
22.04.2024	00:00	PH24042207	1,720.10	9.00
22.04.2024	00:00	PH24042207	1,720.00	30.00
22.04.2024	00:00	PH24042207	1,690.20	14.00
22.04.2024	00:00	PH24042207	1,690.10	570.00
22.04.2024	00:00	PH24042207	1,690.00	200.00
22.04.2024	00:01	PH24042204	1,500.10	20.00
22.04.2024	00:01	PH24042204	1,500.00	35.00
22.04.2024	00:01	PH24042204	1,500.10	20.00
22.04.2024	00:01	PH24042203	1,128.80	1.00

Figure 6. Matched offers in the Intraday Market for 22.04.2024

When the Intraday Market data is grouped by the contract names, thousands of rows are displayed for each contract hour. Since there are thousands of matches occurring per contract hour, it is more convenient to represent them in a price versus cumulative demand curve. In Figure 7, the cumulative demand curve of the Intraday Market matchings at 16.02.2024 12.00 is given. It is plotted by filtering the list of matchings by contract name PH24021612, sorting by decreasing price, and calculating the cumulative sums in the quantity column. The three points marked in red represent quantiles of the cumulative amount. The most left point represents the 25% quantile, which corresponds to a price of 1305 ₺ and an amount of 5664 MWh. The interpretation of this coordinate should be as follows: For the 12 o'clock contract on 16.02.2024, a maximum amount of 5664 MWh could be sold in the Intraday Market with a price offer of 1305 ₺. The remaining two points correspond to the 50% and 75% slices, respectively. On those points, prices are lower and the quantities are higher since there are more buyers for lower

prices. Finding those quantiles in the Intraday Market data will be crucial for the deterministic approach of this study.

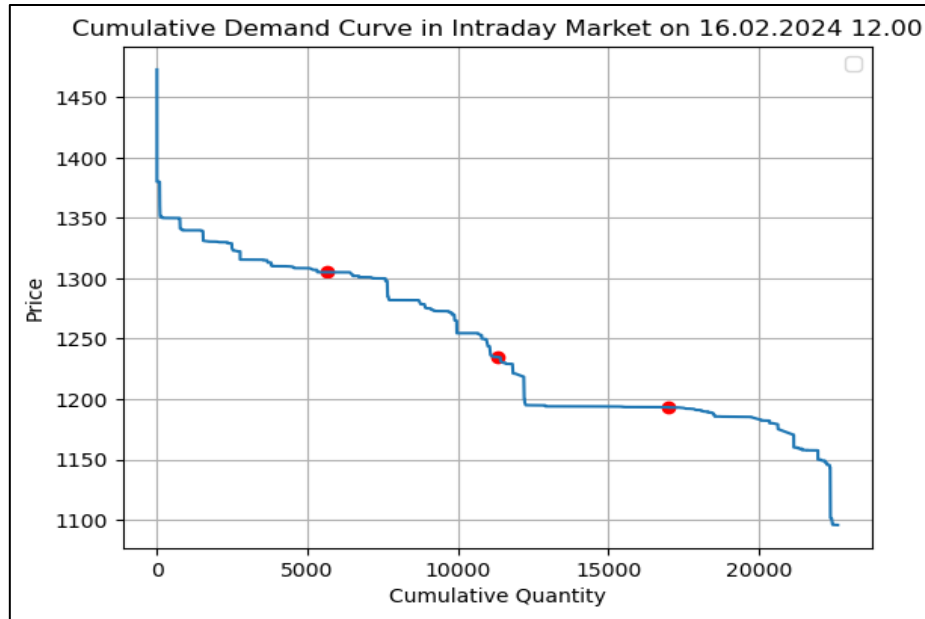


Figure 7. Cumulative demand curve and three quantiles of the Intraday Market for PH24021612 contract

b. Data Analysis

Although data has been available from 2011, data between 2021-2024 was utilized in this study. The reason is to capture the most current market dynamics and avoid the Covid-19 period which affected electricity prices significantly. However, the Russia-Ukraine war, which started in February 2022, also affected electricity prices in the period used in this study. In Figure 8, the plot of the hourly market clearing prices in Turkish Liras in the Day-Ahead Market from 2021 to 2024 is shown. It can be seen that prices increased sharply from 2022 to 2023 due to the energy crisis caused by the war. Apart from this, it can be seen that prices are fixed at certain limits in the upper parts of the chart. This is because the market operator EXIST sets an upper limit for the electricity prices in the offers given to the market. The electricity ceiling price increased until 2023, then the market intervened and electricity prices decreased by lowering the ceiling prices.

Since the deviation of electricity prices in the Turkish lira is greater, plots for the market clearing prices in euros and dollars were also examined. Figure 9 displays the time series

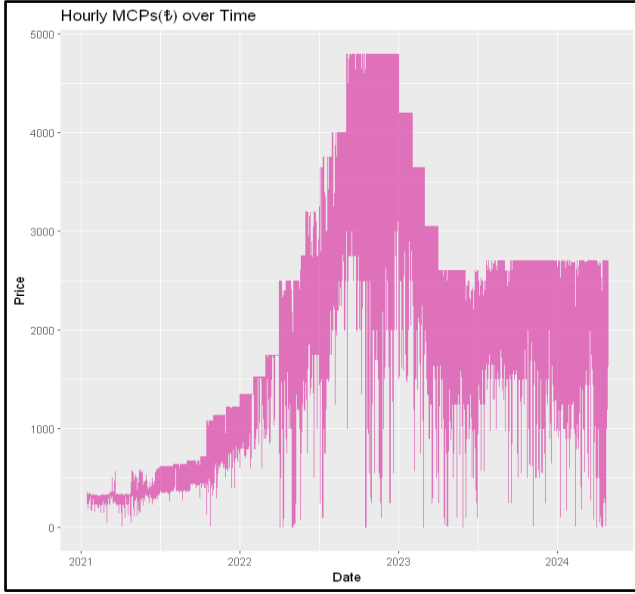
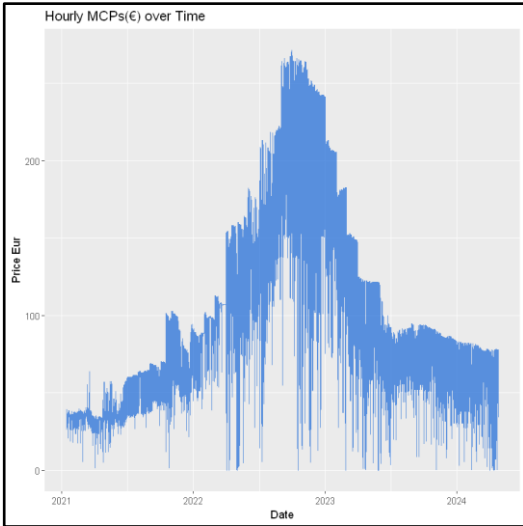
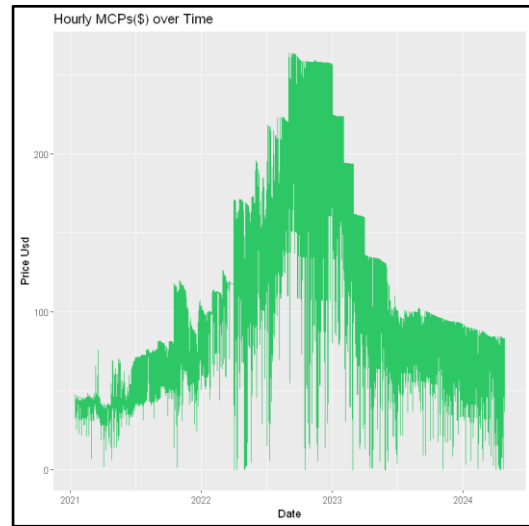


Figure 8. The time series plot of hourly MCPs in ₺

plot for hourly MCPs in euro and dollar, respectively. The increase until 2023 can also be observed in these plots. After 2023, although prices did not change much in Turkish lira, they decreased in euro and dollar because of increasing exchange rates. The ceiling price provision can also be seen in these graphs.



(a) The time series plot of hourly MCPs in €



(b) The time series plot of hourly MCPs in \$

Figure 9. Time series plots of hourly MCPs in different currencies

Next, an autocorrelation function (ACF) is plotted to examine the day-ahead market data in more detail. In Figure 10, the ACF of hourly market clearing prices is shown up to lag 500. As can be seen, the autocorrelation function clearly peaks at multiples of 24, corresponding to a day. In addition, it peaks a bit higher in the multiples of 168, which corresponds to a week. Each data point is interpreted to be highly correlated

with the previous day and the previous week. Therefore, it can be concluded that the market clearing prices have strong daily and weekly seasonalities.

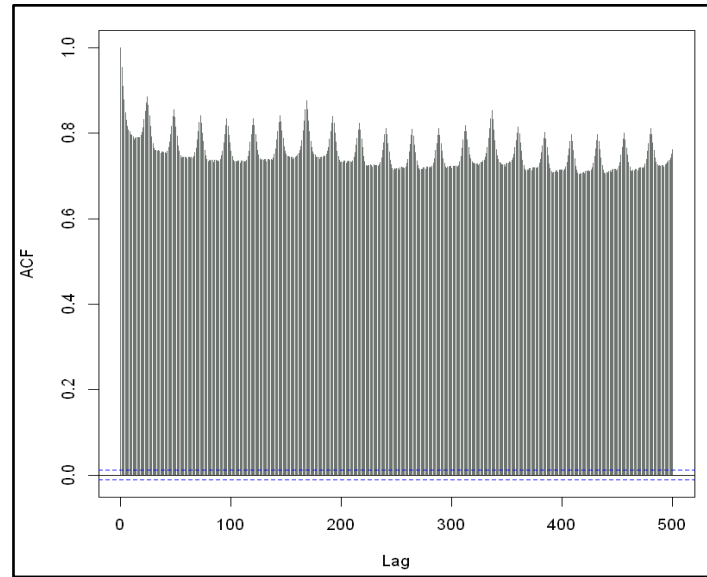
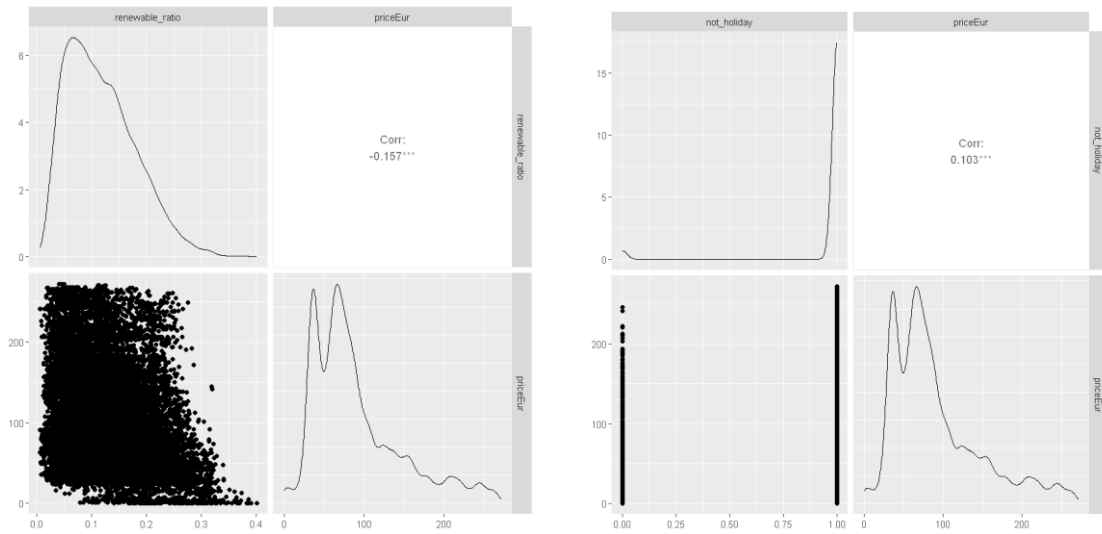


Figure 10. The Autocorrelation Function (ACF) of hourly MCPs

Considering the external factors that may influence electricity prices, the amount of renewable energy production during the day becomes prominent. As explained before, the marginal cost of renewable energy production is particularly low. For this reason, manufacturers can submit their offers at very low prices. When the number of offers at the lower price increase, the supply curve shifts downward and the market clearing price decreases. In this sense, prices are expected to be lower when the share of renewable energy production in total production increases. A new variable has been created to capture this relationship. This variable, called the renewable ratio, is calculated by dividing the solar and wind energy production in a certain hour by the entire production. Figure 11(a) demonstrates the relationship between the renewable ratio and price in euros. They have a slightly negative linear correlation coefficient which is equal to -0.157. In the bottom left chart, it is seen that prices decrease as renewable ratio increases.

Another external factor that affects electricity prices is public holidays. Many factories and industrial plants shut down or reduce operations during public holidays. Since industrial consumers are typically large consumers of electricity, their reduced demand significantly impacts the overall electricity consumption. Electricity supply from

power plants (including renewable sources like wind and solar) generally remains consistent regardless of holidays. Since the supply does not decrease proportionately with the drop in demand, the market experiences an excess of supply, resulting in lower prices. The holiday information is added to the Day-Ahead Market data as a categorical variable. In the bottom left chart of Figure 10(b), it is observed that the prices are lower when it is a holiday. The average hourly price when it is not a holiday is calculated as €91.21, while it is only €61.01 when it is a holiday.



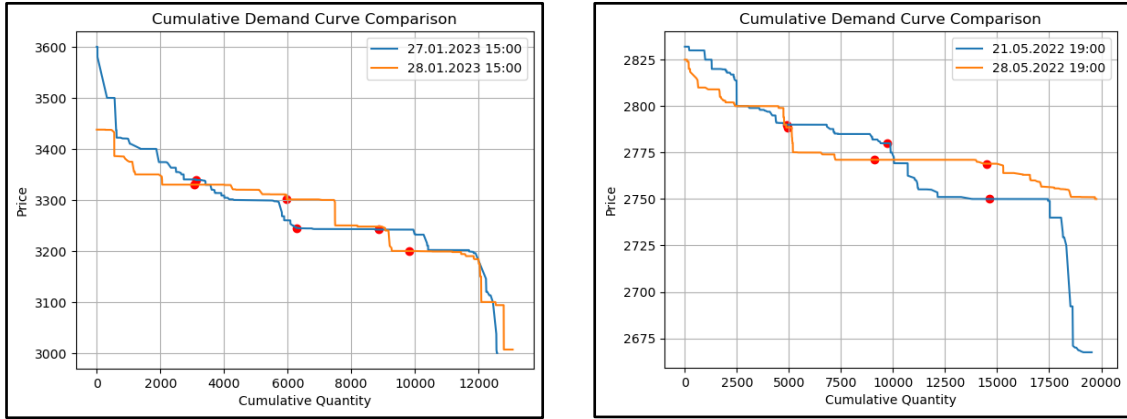
(a) Relationship between renewable ratio and prices in euros

(b) Relationship between public holidays and prices in euros

Figure 11. Pairwise plots of external variables and prices in euros

It is much more difficult for the Intraday Market to predict volume and price and attribute them to external factors. However, the relationship between the Day-Ahead Market and the Intraday Market may be anticipated. When the Day-Ahead Market is not sufficient to satisfy the demand, transactions continue in the Intraday Market to balance the market. Therefore, the Day-Ahead and the Intraday markets data at a certain contract hour are expected to be consistent with each other. Since the Intraday Market operates based on bilateral matchings, the volume and price dynamics of this market especially depend on the participants' own preferences. The challenge of foreseeing the dynamics of the Intraday Market, therefore, remains.

The strong daily and weekly seasonalities seen in the Day-Ahead Market may be a sign that such a relationship can also be seen in the Intraday Market. For comparison, cumulative demand curves of the Intraday Market for consecutive days and weeks are plotted. In Figure 12, cumulative demand curves together with the three quantile points of the same contract hour for consecutive days and consecutive weeks are shown.



(a) Cumulative demand curves for contract hour 15.00 in 27.01.2023 and 28.01.2023

(b) Cumulative demand curves for contract hour 19.00 in 21.05.2022 and 28.05.2022

Figure 12. Cumulative demand curve comparisons for consecutive days and consecutive weeks

As a result, while the market clearing price can be associated with external factors and thus may be predicted, predicting intraday market dynamics alone is a complex problem in itself. However, the fact that these markets contain daily and weekly seasonality may indicate that the answer may be in the past when making predictions about the market dynamics of the next day. In other words, if the next day can be compared to a day in the past in certain respects, it can be said that market dynamics will also be similar to this day. This is the main idea of the prediction models established in the following sections.

5. DESCRIPTION OF DERIVED MODEL ALTERNATIVES

In this project, four distinct predictive approaches are derived, implemented, and analyzed from the deterministic modeling approach to predict the market prices and optimize bidding strategies for the next day. These approaches include the Day-Before model, Week-Before model, Rule-Based model, and K-Nearest Neighbor (KNN) model.

Firstly, to implement these approaches, a deterministic model is created which describes Turkish electricity market dynamics. In what follows, the notation and the mathematical model are provided:

Sets and Indices

- t, T : Set of hours in a day, $t=0,1,2,\dots,23$
- q, Q : Set of quantiles, $q=25, 50, 75$

Decision Variables

- $Q1(t)$: Quantity offered in the Day-Ahead Market for contract hour t
- $Q2(t,q)$: Quantity offered in the Intraday Market for quantile q for contract hour t

Parameters

- $MCP(t)$: Market clearing price per MWh for contract hour t in the Day-Ahead Market.
- $IDP(t,q)$: Price offered in the Intraday Market for quantile q for contract hour t .
- $P(t)$: The total production amount in hour t
- $D(t,q)$: The maximum demand in the Intraday Market for quantile q for contract hour t

$$\max \sum_t (MCP(t) \cdot Q1(t) + \sum_q IDP(t,q) \cdot Q2(t,q)) \quad (1)$$

subject to

$$Q1(t) + \sum_q Q2(t,q) \leq P(t) \quad \forall t \quad (2) \text{ Production/Capacity Constraint}$$

$$Q2(t,q) \leq D(t,q) \quad \forall t, q \quad (3) \text{ Demand Constraint}$$

$$Q1(t), Q2(t,q) \geq 0 \quad \forall t, q \quad (4) \text{ Cardinality Constraints}$$

In the deterministic model, the objective function (1) aims to maximize the revenues from two markets: the Day-Ahead Market and the Intraday Market.

This first constraint (2) is the production capacity. This constraint describes the fact that the total amount offered to the Day-Ahead Market and the Intraday Market cannot exceed the total production amount for a given time t .

The second constraint (3) is related to how the Intraday Market dynamics are described. To estimate the respective prices in the Intraday Market, the cumulative demand curve is divided into quantiles. Three different quantiles (25, 50, and 75) are used to distribute the risks. As can be seen from the given demand curve realization below, once the quantile points are found, related price and demand estimations can be received from the curve.

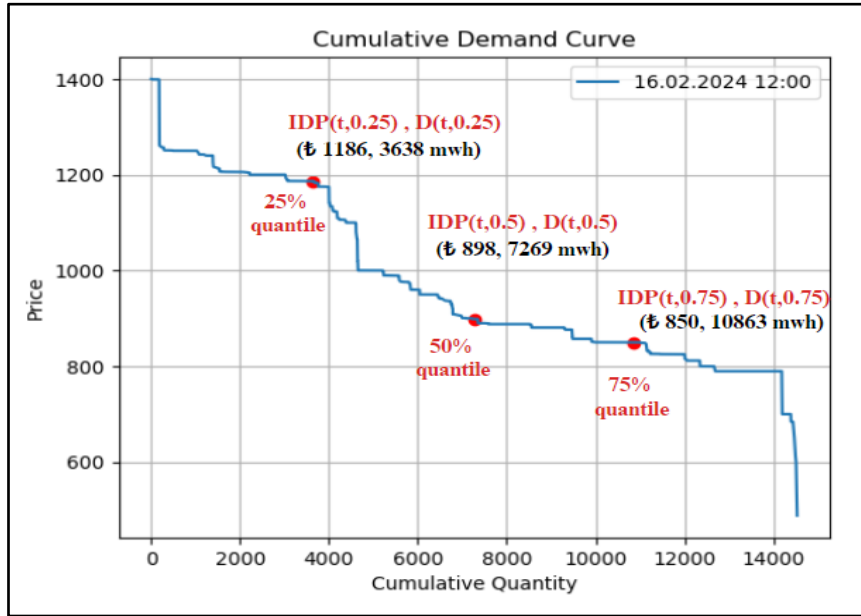


Figure 13. Derivation of $D(t,q)$ and $IDP(t,q)$ from cumulative demand curve

For exemplifying, consider the 25% quantile point on the curve. The price associated with the point implies that the 25 % of the total demand in the Intraday Market is sold at a higher or equal to this price. The offered prices in the Intraday Market,

IDP(t, q), will be determined by the associated price coordinate of the respective quantile in this way.

Once the quantiles and their respective prices are determined, the information on the maximum amounts demanded at those prices can also be found. The parameter of the maximum demand $D(t, q)$ used in constraint (3) is determined as such. The constraint (3) sets an upper limit on the decision variable $Q2(t, q)$ which is the amount that will be bid on the Intraday Market for quantile q for contract hour t .

Lastly, the cardinality constraints (4) for the decision variables, $Q1$ and $Q2$ are added to the model.

The provided deterministic model maximizes the sum of revenues from both the Day-Ahead Market and the Intraday Market across all hours of the day and all quantiles. This framework provides a comprehensive approach to optimize bidding strategies by considering both markets simultaneously.

a. Assumptions and Limitations

As discussed earlier, the problem is solved from a small-sized renewable energy producer's point of view. The production level for each hour for the next day, $P(t)$, is assumed to be certain and given. This assumption does not account for potential fluctuations in renewable energy production due to weather conditions or technical issues. Historical data for production levels is taken from the Bergres Elektrik Üretim A.Ş. In the Day-Ahead Market, offers are assumed to be given at zero price to ensure the offered quantity is accepted. Although bidding at zero price may shift the supply curve downwards, it is assumed that the offers do not influence the prices. Since the production levels of the small-sized producer may be considered insignificant in the overall supply, it is safe to assume that the producer is a price-taker in the Day-Ahead Market.

For the Intraday Market, among the different types of bids, only hourly bids were considered in the historical data. Although bids can be given up to one hour in advance of the physical delivery in the intraday market, it is assumed that the allocation problem is solved the day before and the decision is not changed later ignoring the dynamic nature of the Intraday Market where conditions can change rapidly, and real-time adjustments

are often necessary. In cases where the bid could not find a buyer and therefore could not be sold, the problem is formulated as if the remaining energy does not bring any cost and the producer does not participate in the balancing power market to trade the excess energy. In reality, producers often trade excess energy in balancing markets, which could impact overall profitability and operational strategies. In the Day-Before Model, it is assumed that all Intraday Market transactions for today are completed as soon as the problem is solved for the next day.

By acknowledging the limitations that are led by the assumptions that are made, the model can be better understood in terms of its applicability and potential areas for improvement. Future work could focus on addressing these limitations to enhance the model's accuracy and relevance in more complex and dynamic market conditions.

b. Day-Before Model

The Day-Before model is a simplistic yet effective method for predicting market prices based on the assumption that tomorrow's prices will be identical to today's. The daily seasonality seen in both markets forms a basis for the Day-Before model. This model leverages the immediate past to forecast future prices using the large correlation observed between consecutive days.

In this model, $MCP(t)$ for each hour t is set equal to the price observed in the same hour of the previous day. For the Intraday Market, the demand quantities and related prices are also taken from the observed market of the previous day, similarly. The quantities $Q1(t)$ and $Q2(t,q)$ are then optimized based on these predicted prices. The results in terms of the revenues will be discussed in the following sections.

c. Week-Before Model

The Week-Before model predicts that the prices on a given day will mirror the prices observed on the same day of the previous week. This approach accounts for the weekly seasonality patterns observed.

In this model, $MCP(t)$ for each hour t is set equal to the price observed in the same hour of the corresponding day of the previous week. Similarly, the Intraday Market is

assumed to be the same as the observed market of the previous week. The quantities $Q1(t)$ and $Q2(t,q)$ are optimized based on these predicted prices. Resulting revenues of the Week-Before model will be shared in the following sections.

d. Rule-Based Model

The Rule-Based model combines predictive modeling and historical data patterns. The main idea of the rule-based model, following the day-before and week-before models, is as follows: tomorrow can be estimated by the most similar day in the past, which will resemble tomorrow in some features. Since the market clearing prices can be estimated using predictive algorithms, the first step of the Rule-Based Model is to forecast the market clearing prices in the Day-Ahead Market for the next day. Several different algorithms, such as time series regression and ARIMA, are experimented with, and the model with the fewest error measures is chosen.

	MODEL	RESPONSE	PREDICTORS	MSE	MAE	MAPE	WMAPE
1	ARIMA (2,0,2)	Daily Difference (€)	Holiday, Holiday(-24)	789.96	15.14	37474.5	0.1681
2	ARIMA (0,0,0)	Daily Difference (€)	Holiday, Holiday(-24), Hour	738.3	15.18	37461.1	0.1685
3	ARIMA (2,0,2)	Daily Difference (€)	Holiday, Holiday(-24), IS_WAR	739.3	15.14	37449.1	0.1681
4	T.S. Regression	Daily Difference (€)	Difference(-24), Difference(-168), Holiday, Holiday(-24), IS_WAR	592.5	14.17	39253.08	0.160
5	T.S. Regression	Daily Difference (€)	Difference(-24), Difference(-168), Holiday, Holiday(-24), Hour	592.4	14.47	39250.5	0.1601
6	T.S. Regression	Price (€)	Price(-24), Price(-168), Holiday, Holiday(-24), Holiday(-168)	500.1	13.75	39405.2	0.153
7	T.S. Regression	Price (€)	Price(-24), Price(-168), Holiday, Holiday(-24), Holiday(-168), renew_ratio	186.3	10.39	24741.6	0.152

Figure 14. Predictive algorithms and the error measures for estimating market clearing prices

A time series regression model is utilized to predict the prices using several exogenous variables such as autoregressive terms, holiday information, and the renewable ratio. The renewable ratio variable for the next day was taken as a forecast from an external source in the forecasting phase.

With the market clearing price estimate in hand, the most similar day that will estimate the next day is found using historical data. For each 23 contract hour of the next day, historical data points belonging to the same contract hour are searched. The ones

having the same holiday information and the market clearing price within 15% of the next day's predicted price are identified. After finding those data points for each hour, a voting algorithm is utilized to find the most similar day. The number of occurrences of each day in the set of similar data points for all contract hours is counted, and the most similar day is selected as the most prevalent one.

For the optimization model parameters, the market clearing price predictions are used directly for $MCP(t)$. For the Intraday Market, demand and price parameters are assumed to be the same as those of the identified similar day. Then the optimization model is solved for the allocation between two markets.

e. K-Nearest Neighbor (KNN) Model

The KNN model predicts the next day's market prices for a contract hour by finding the three nearest neighbors for each hour from historical data. The distances between the target data points and historical data points are calculated by using the renewable ratio prediction for tomorrow, production forecasts, yesterday's prices, last week's prices, and the holiday information (whether it is a holiday tomorrow or not). Then, after the scaling of these features, the nearest neighbors are calculated.

For each contract hour, three nearest neighbors are found. Then, the model identifies the most recurring date among these 72 data points (24 hours * 3 neighbors), and the predictor date is chosen as the most recurring one. Both Day-Ahead and Intraday Market prices are assumed to match those of the selected date.

In this model, $MCP(t)$ and $IDP(t,q)$ for each hour t are set equal to the prices observed on the most recurring date identified from the nearest neighbors. Quantities $Q1(t)$ and $Q2(t,q)$ are then allocated optimally based on these predicted prices.

6. COMPARISON OF ALTERNATIVES AND FURTHER ANALYSIS

These four models offer a range of approaches from simple historical methods to more complex predictive techniques, allowing for flexibility in optimizing bidding strategies for renewable energy portfolio management in electricity markets. Each model

leverages different aspects of historical and predictive data to maximize revenues in both the Day-Ahead and Intraday Markets.

Before comparing the different approaches and resulting revenues, the concept of expected and realized revenues shall be mentioned. When the model is solved for the next day, it uses the predicted parameters. However, the optimal objective value may not be realized when the true parameters are revealed. When the real market clearing price is determined, the Day-Ahead return in the objective function will be different than what is found in the optimal solution, i.e., expected. Similarly, the quantity offered in the Intraday Market may not find a buyer for the offered price. In this case, the transaction would not be made and the expected revenue from the Intraday Market cannot be realized. Therefore, every time the model is solved, the actual parameters must be checked and the real revenues of the decisions must be calculated.

To implement the four models, the Python programming language is used and the Pulp package is utilized as a solver. The R programming language is also used for building forecasting models for market clearing prices.

For comparison purposes, the deterministic model is solved for a year from April 2023 to April 2024 with the true parameters, assuming each next day is perfectly known from the previous day. The resulting annual revenue is calculated as ₦455,702,712 in the best case. This will set an upper bound for the performances of four approaches. The percentage of realized revenues for each model relative to the upper limit will determine the overall success of the models.

a. Overall Performances

After solving the four described models for a period of one year, the results were as follows:

The Rule-Based model, utilizing discrete rules to estimate parameters, demonstrated the highest performance among the examined models, achieving total revenue amounting to 73% of the optimum case, i.e., upper limit. The Rule-Based model, therefore, is the most effective model in terms of revenue generation.

The second-best model was the Day-Before model, with only a slight difference in performance compared to the Week-Before model. Both models showed relatively similar results, indicating their comparable effectiveness in predicting market prices. The Day-Before model achieved 72.3% of the optimum case, while the Week-Before model reached 72.1% of the optimum case.

The K-Nearest Neighbor (KNN) model, however, exhibited significantly poorer performance compared to the other models. This model achieved only 63.3% of the upper bound in terms of revenues. A deeper analysis revealed that this discrepancy was primarily due to the Intraday Market. Despite the expectation that the KNN model would perform better by leveraging multiple data points, its actual performance was prevented by the chosen nearest neighbors.

A notable observation was that the KNN model frequently selected the previous day as the nearest neighbor, similar to the Day-Before model. This suggests that the features used for finding nearest neighbors often had values very close to those of the previous day, leading the KNN model to operate like the Day-Before model for much of the year.

However, there were instances where the KNN model selected different dates throughout the year. These selections did not perform as efficiently, resulting in lower total revenue compared to the Day-Before model. This indicates that while the KNN model has the potential for better performance with diverse data points, its current implementation did not consistently outperform simpler models in this context. Figure 15 compares the overall performances of the four models with the optimum case in terms of revenues.

b. Performances According to the Markets

The Rule-Based model demonstrates strong performance in the Day-Ahead Market, achieving the highest revenue among the models examined. In the Intraday Market, it delivers similar revenue levels to the Week-Before and Day-Before models.

In contrast, the K-Nearest Neighbor (KNN) model is less efficient in both the Day-Ahead and Intraday Markets compared to the other models. Its performance is notably lower, highlighting its relative inefficiency in predicting market dynamics.

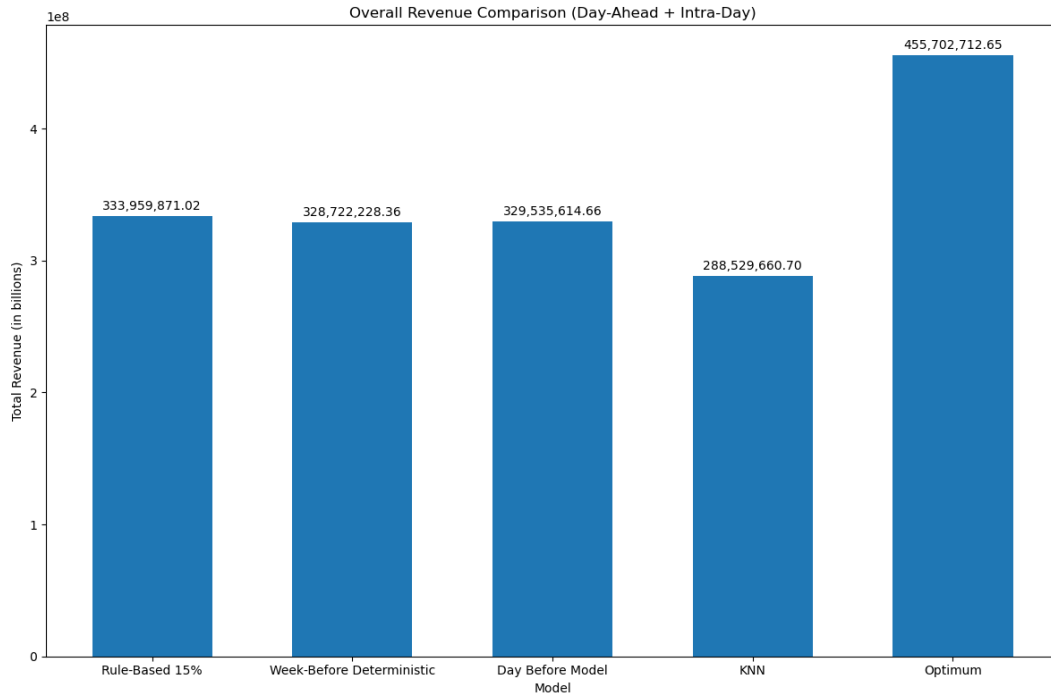


Figure 15. Total realized revenue comparison of models

Figure 16 illustrates the overall performances for the Day-Ahead and Intraday Markets separately for each model. The optimal case clearly shows that the Intraday Market is highly profitable. However, as evidenced by the results, predicting the dynamics of the Intraday market is significantly more challenging. The models show varying degrees of success, with the Rule-Based model being the most effective overall, yet still facing difficulties in capturing the complex nature of the Intraday market.

These findings underscore the necessity for more sophisticated and nuanced approaches to better predict and optimize performance in the Intraday market, which remains a critical component for maximizing overall revenue.

c. Decisions on Allocation Percentages

The allocation of production between two markets demonstrates that the models aim to leverage the benefits of the Intraday Market by allocating approximately 60% (and up to 65% in the KNN model) of production to this market. However, as indicated in the revenue comparisons, the models struggle to accurately predict the related prices and quantities in the Intraday demand, leading to lower revenues. Despite the higher

allocation to the Intraday market, the models' inability to capture the market dynamics effectively results in suboptimal performance in terms of revenue generation.

This highlights the complexity and challenges associated with predicting Intraday Market conditions accurately and underscores the need for more sophisticated modeling approaches to improve predictions and optimize bidding strategies.

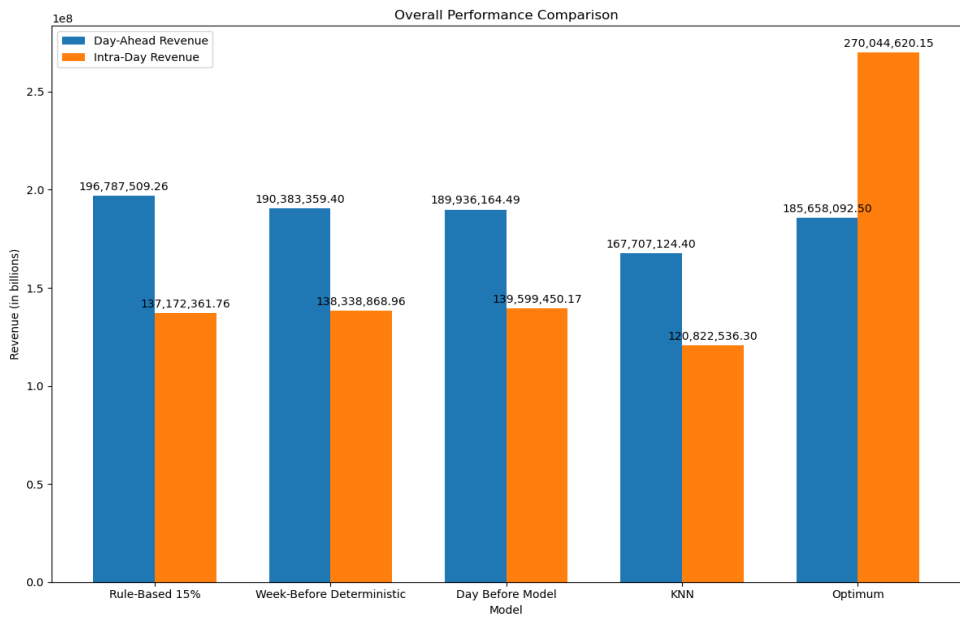


Figure 16. Market-based realized revenue comparison of models

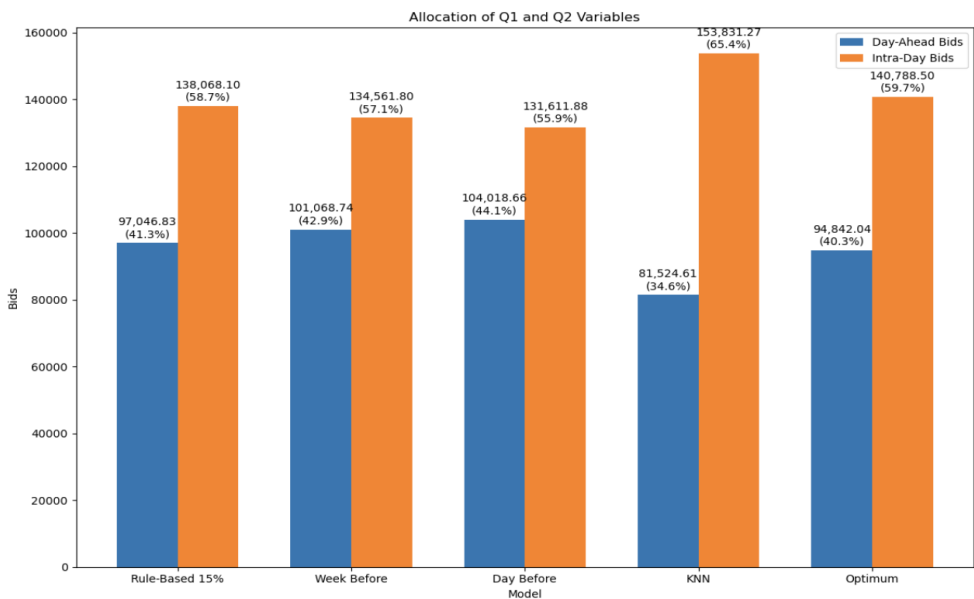


Figure 17. Allocation of the production comparison of models

d. Seasonal Performances

The chart below displays the monthly total revenue comparison (combining Day-Ahead and Intraday revenues) across the four models and the optimal case. All models follow a similar pattern to the optimal case throughout the year.

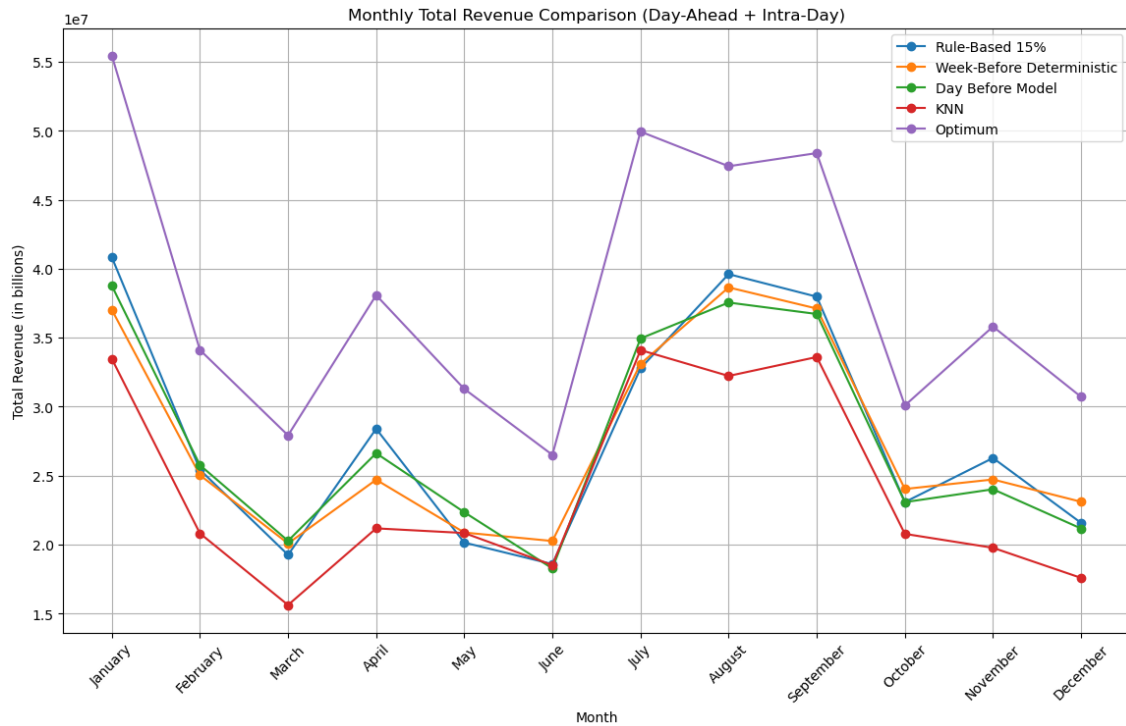


Figure 18. Monthly total realized revenue comparison of models

Key Observations:

1. Pattern Similarity:

All models exhibit a similar revenue pattern to the optimal case, indicating consistent seasonal trends across the year.

2. Intraday Market Challenges:

Despite following the optimal pattern, all models generate significantly lower revenue compared to the optimal case. This discrepancy is primarily due to the models' inability to accurately predict the dynamics of the intraday market.

The best-performing model still struggles to correctly forecast intraday prices and related demand amounts, resulting in lower revenues. (Appendix A & Appendix B)

3. ANOVA Analysis:

To determine if there is a statistically significant difference between the models' performances in the Intraday and Day-Ahead Markets, ANOVA was used.

After calculating and checking the within-group and between-group variances for the models over 12 months, no statistically significant difference was found between the models in either market (F-statistic: 0.576, p-value: 0.633 for Day-Ahead; F-statistic: 0.418, p-value: 0.741 for Intraday).

This finding suggests that all models perform similarly in both markets, despite their varying approaches.

Given the consistent challenges in accurately predicting intraday market dynamics, it was decided to revise the deterministic approach, particularly in handling the complex and hard-to-predict intraday demand curves. This adjustment aims to enhance the models' predictive accuracy and overall performance, aligning more closely with the optimal case.

7. ANOTHER APPROACH: SCENARIO-BASED STOCHASTIC OPTIMIZATION

From the analysis of the other four models, it became evident that while the Intraday Market is quite profitable, the previous quantiles method could not adequately capture the dynamics of this market. The scenario-based approach was adopted to address this limitation and improve the accuracy and robustness of the predictions.

Key Changes in the Scenario-Based Approach:

1. Generation of Scenarios:

The scenario-based optimization model builds upon the deterministic approach by incorporating multiple scenarios to account for the uncertainties and variability in the market.

This model utilizes a similar method to the K-Nearest Neighbor (KNN) model for generating different scenarios by finding nearest neighbors but extends it to consider a broader set of possible future outcomes when solving the allocation problem.

2. Weighted Average Price for Intraday Market (IDP):

For the intraday market price $IDP(t,s)$, the model now uses the weighted average price information for the given hour from the related day's market data.

This change aims to provide a more accurate representation of the market prices by considering the variability and trends observed in the historical data.

3. Demand Calculation:

For each scenario, the demand $D(t,s)$ is calculated based on the bilateral agreements completed on that day's intraday market at prices higher than the given weighted average.

By summing all the quantities from these agreements, the model captures the total demand quantity at the specified price, offering a more realistic estimate of market demand.

4. Introduction a Variable for Sales, $S(t,s)$:

A new variable $S(t,s)$ is introduced, describing the sales amount in each different scenario s , at each different hour t . Then, this variable is connected to the related demand on each scenario s and time t , and the total allocation amount decided, $Q2$.

Benefits of the Scenario-Based Optimization Model

- **Improved Accuracy:** By considering multiple scenarios and using weighted average prices, the model provides a more precise prediction of market dynamics.
- **Flexibility:** The scenario-based approach can adapt to different market conditions and better handle the inherent uncertainties in market prices.
- **Comprehensive Analysis:** This model offers a holistic view of potential future outcomes, enabling more informed decision-making and strategic planning.

Model Details:

1. Scenario Generation:

- For each hour t , the model identifies the 10 nearest neighbors from historical data based on market prices.
- This results in a set of 240 data points (24 hours * 10 neighbors).
- From these 240 data points, the most recurring 10 dates are selected, each representing a unique scenario s .

2. Objective Function:

- The objective is to maximize the average profits across all scenarios by solving for the quantities $Q1(t)$ and $Q2(t)$.

$$\max \quad 1/|S| \sum_s \sum_t (MCP(t,s) \cdot Q1(t) + IDP(t,s) \cdot S(t,s))$$

where S is the set of scenarios, $|S|$ is the number of scenarios, $MCP(t,s)$ is the market clearing price for scenario s at hour t , and $IDP(t,s)$ is the Intraday price for scenario s at hour t .

3. Constraints:

$$Q1(t) + Q2(t) \leq P(t) \quad \forall t \quad (1) \text{ Production/Capacity Constraint}$$

Similarly to the other models, this first constraint is for the hourly production capacity.

$$S(t,s) \leq D(t,s) \quad \forall t,s \quad (2) \text{ Sales Constraint}$$

This new sales constraint, bounds the sales variable into taking values smaller than the total demand in each different hour, in each different scenario. Meaning that an amount of electricity can be sold is bounded by the given demand.

$$S(t,s) \leq Q2(t) \quad \forall t,s \quad (3) \text{ Sales Constraint}$$

The second sales constraint is for bounding the Q2 variable to be larger than all sales values, in each scenario. By this constraint, only one value for the Q2 decision variable is determined for the allocation problem, and the possible sales information across scenarios is also stored on the S(t,s) variables.

$$Q1(t), Q2(t,q) \geq 0 \quad \forall t, q \quad (4) \text{ Cardinality Constraints}$$

Lastly, these constraints are the cardinality constraints for the decision variables, Q1 and Q2.

In summary, the scenario-based optimization model enhances the predictive accuracy and robustness of the bidding strategies, making it a valuable addition to the other models for renewable energy portfolio management in electricity markets.

8. CONCLUSIONS AND DISCUSSION

a) Summary of Deterministic Model Approaches

This study investigated various models to optimize bidding strategies for a renewable energy portfolio in electricity markets. The models examined include the Day-Before model, Week-Before model, Rule-Based model, K-Nearest Neighbor (KNN) model, and an advanced Scenario-Based Optimization model. The following table provides a summary of the findings and discusses the implications of the results.

Model	Expected Annual Revenue	Real Annual Revenue	Percentage of Optimal Case
Day-Before Model	£472,371,007	£329,535,614	72.3%
Week-Before Model	£479,220,256	£328,722,228	72.1%
Rule-Based Model	£473,290,415	£326,117,383	71.6%
K-Nearest Neighbor (KNN) Model	£700,435,206.38	£288,529,660	63.3%

Figure 19. Summary of findings

These models demonstrated consistent performance in the Day-Ahead Market; however, the Intraday Market presented significant challenges. The previous quantiles method did not adequately capture the Intraday Market dynamics, resulting in lower revenues in real scenarios.

b) Summary of Enhanced Scenario-Based Stochastic Optimization Model

To address the limitations observed in the earlier models, a new Scenario-Based Optimization model was introduced. This model employs a more detailed approach to describing the Intraday Market by considering weighted average price information and bilateral agreements with prices higher than the weighted average.

Metric	Value
Optimal Case Revenue	£4,251,211,848
Real Revenue	£4,197,395,963
Percentage of Optimal Case	98.73%

The new methodology significantly improved the understanding and prediction of the Intraday Market, resulting in a much higher optimal case revenue. Although the Scenario-Based Model did not achieve the optimal case, it managed to capture 98.73% of the optimal case revenue, indicating a marked improvement over the previous models.

c) Overall Conclusions

Performance of Models: The Day-Before, Week-Before, and Rule-Based models performed consistently, achieving around 71-72% of the optimal case revenues. The KNN model, despite its sophisticated approach, underperformed in real scenarios with only 63.3% of the optimal case revenue. In the Scenario-Based Model, 98.73% of the optimal case revenue is achieved, resulting in the most efficiently working model with the increased optimal revenue as well, due to re-defined Intraday Market dynamics calculations.

Intraday Market Dynamics: The Intraday Market has proved to be a significant challenge in the initial models. The quantiles method failed to capture the complexities, leading to lower real revenues. The new Scenario-Based Optimization model effectively addressed these shortcomings.

Improved Methodology: The enhanced approach for the Intraday Market in the scenario-based model led to a substantial increase in the optimal case revenue. The new model's real revenue performance at 98.73% of the optimal case demonstrates its potential for better capturing market dynamics and improving bidding strategies.

d) Implementation and Discussions

The proposed solution involves optimizing bidding strategies using a scenario-based optimization model. This model leverages historical data and machine learning techniques to improve revenue predictions. However, this model involves some limitations as it is assumed to have zero market costs for simplicity. Also, the model relies on accurate historical data, which may not always reflect future conditions. When it comes to the sensitivity of the model, it should be noted that this model may require frequent updates to reflect changing market dynamics, such as upper limits on electricity prices determined by authorities.

Overall, the solution model aligns with system requirements in terms of timeline, addressing the need for accurate market predictions and efficient bidding strategies. It can be implemented using existing data analysis tools and optimization algorithms. The model is robust, but its performance depends on forecast accuracies of MCP and other variables as well as market stability. However, the solution of all proposed models requires regular updates and maintenance. It needs continuous data collection and management to ensure accuracy. The solution should be revised frequently, ideally before 12:30 noon daily, before the Day-Ahead Market closes. Regular reviews and adjustments based on market trends and new data are crucial for sustained performance.

To implement the proposed methodologies, several industrial engineering skills, tools, and techniques were essentially used. Operations research techniques, such as linear programming and optimization algorithms, were pivotal in solving the allocation problem and maximizing revenues across different market scenarios.

Statistical analysis and data analytics tools, including time series regression analysis and ANOVA, were utilized to analyze historical data, predict market trends, and validate model performance. Additionally, machine learning algorithms, as used in

nearest neighbor searches, enhanced the accuracy of market predictions. Lastly, the use of software tools such as Python and R facilitated the development, testing, and implementation of the proposed models. Integrating these industrial engineering skills and techniques provided a comprehensive and effective approach to optimizing renewable energy bidding strategies in electricity markets.

Economically, the design maximizes revenues for renewable energy producers, fostering financial sustainability. Environmentally, it promotes the use of renewable energy, reducing reliance on fossil fuels and lowering carbon emissions. Ethically, it ensures fair market participation by treating small producers as price-takers. Societally, it supports the transition to clean energy, contributing to a more sustainable and resilient energy system, which benefits communities by providing more stable and eco-friendly energy sources.

For further studies on this problem, supply chain management principles may aid in understanding and managing the flow of energy from production to market, ensuring efficiency and minimizing losses. Additionally, the application of simulation techniques may help model and understand the complex dynamics of the Intraday and Day-Ahead Markets, allowing for better scenario planning and decision-making.

In conclusion, this study demonstrates that a detailed and nuanced approach to modeling market dynamics, particularly for the Intraday Market, is crucial for optimizing renewable energy portfolio management in electricity markets. The Scenario-Based Optimization Model represents a significant step forward, offering a robust framework for achieving higher revenues and more effective bidding strategies.

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APPENDIX

APPENDIX A

Visit the [GitHub link](https://github.com/batuoazkn/IE492-ElectricityMarkets) for the source codes.

[<https://github.com/batuoazkn/IE492-ElectricityMarkets>]

APPENDIX B

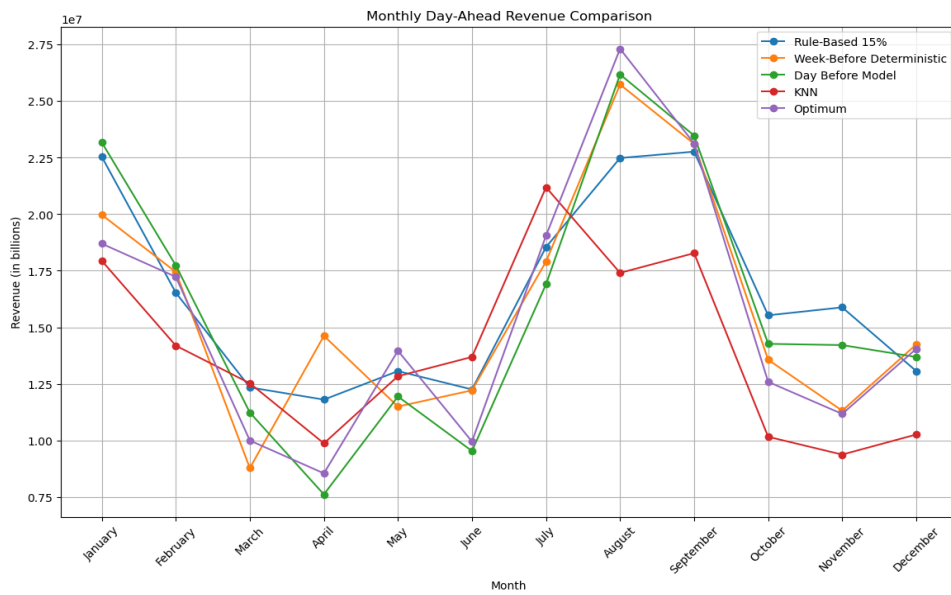


Figure B-1. Monthly revenue comparisons of proposed models in the Day-Ahead Market

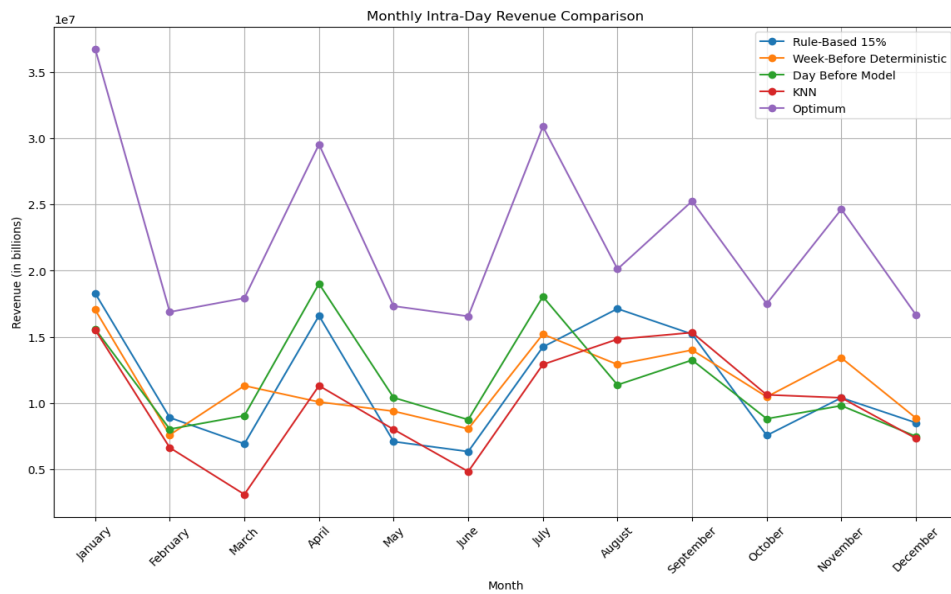


Figure B-2. Monthly revenue comparisons of proposed models in the Intraday Market